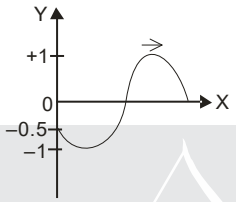
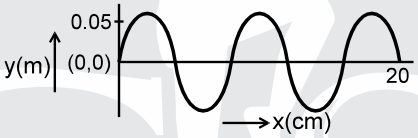
SYLLABUS : STRING WAVE & SOUND WAVE

1. The equation of a wave travelling along the positive x–axis, as shown in figure at t = 0 is given by:–



(A) $\sin\left(kx-\omega t+\frac{\pi}{6}\right)$ (B) $\sin\left(kx-\omega t-\frac{\pi}{6}\right)$ (C) $\sin\left(\omega t-kx+\frac{\pi}{6}\right)$ (D) $\sin\left(\omega t-kx-\frac{\pi}{6}\right)$

2. For the wave shown in figure, the equation for the wave, travelling along +x axis with velocity 350 ms^{-1} when its position at t = 0 is as shown



(A) 0.05 sin $(\frac{314}{4}x - 27475\ t)$ (B) 0.05 sin $(\frac{379}{5}x - 27475\ t)$

(C) 1 sin $(\frac{314}{4}x - 27475\ t)$ (D) 0.05 sin $(\frac{289}{5}x + 27475\ t)$

3. The displacement of a wave disturbance propagating in the positive x-direction is given by $y = 1/(1 + x^2)$ at time t = 0 and $y = 1/[1 + (x - 1)^2]$ at t = 2 seconds where x and y are in meters. The shape of the wave disturbance does not change during the propagation. The velocity of the wave is

(A) 2.5 m/s (B) 0.25 m/s (C) 0.5 m/s (D) 5 m/s

4. A transverse wave is described by the equation $Y = Y_0 \sin 2\pi (ft - x/\lambda)$. The maximum particle velocity is equal to four times the wave velocity if

(A) $\lambda = \pi Y_0/4$ (B) $\lambda = \pi Y_0/2$ (C) $\lambda = \pi Y_0$ (D) $\lambda = 2\pi Y_0$

Power of real gurus

5. A travelling wave on a string is given by $y = A \sin [\alpha x + \beta t + \frac{\pi}{6}]$. If α = 0.56 /cm, β = 12/sec, A = 7.5 cm, then find the position and velocity of particle at x = 1 cm and t = 1s is

(A) 4.6 cm, 46.5 cm s^{-1} (B) 3.75 cm, 77.94 cm s^{-1}

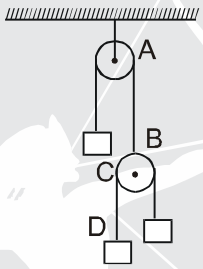
(C) 1.76 cm, 7.5 cm s^{-1} (D) 7.5 cm, 75 cm s^{-1}

6. A transverse wave of amplitude 0.50m, wavelength 1m and frequency 2 Hz is propagating on a string in the negative x-direction. The expression of the wave is

(A) $y(x, t) = 0.5 \sin (2\pi x - 4\pi t)$ (B) $y(x, t) = 0.5 \cos (2\pi x + 4\pi t)$

(C) $y(x, t) = 0.5 \sin (\pi x - 2\pi t)$ (D) $y(x, t) = 0.5 \cos (2\pi x - 2\pi t)$

7. Both the strings, shown in figure, are made of same material and have same cross-section. The pulleys are light. The wave speed of a transverse wave in the string AB is v_1 and in CD it is v_2. The v_1/v_2 is



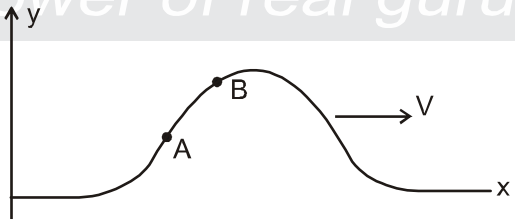
(A) 1 (B) 2 (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

8. Two small boats are 10m apart on a lake. Each pops up and down with a period of 4.0 seconds due to wave motion on the surface of water. When one boat is at its highest point, the other boat is at its lowest point. Both boats are always within a single cycle of the waves. The speed of the waves is :

(A) 2.5 m/s (B) 5.0 m/s (C) 14 m/s (D) 40 m/s

9. A wave pulse is generated in a string that lies along x-axis. At the points A and B, as shown in figure, if R_A and R_B are ratio of wave speed to the particle speed respectively then :

Power of real gurus



(A) $R_A > R_B$ (B) $R_B > R_A$

(C) $R_A = R_B$ (D) Information is not sufficient to decide.

10. The equation of a wave is $y = 4 \sin\left[\frac{\pi}{2}\left(2t + \frac{x}{8}\right)\right]$ where y, x are in cm and time in seconds. The amplitude, wave length, velocity and frequency of the wave are, respectively,

(A) 4 cm, 32 cm, 16 cm/s, 0.5Hz (B) 8 cm, 16 cm, 32 cm/s, 1.0Hz

(C) 4 cm, 32 cm, 32 cm/s, 0.5Hz (D) 8 cm, 16 cm, 16 cm/s, 1.0Hz

11. A loop of a string of mass per unit length μ and radius R is rotated about an axis passing through centre perpendicular to the plane with an angular velocity ω. A small disturbance is created in the loop having the same sense of rotation. The linear speed of the disturbance for a stationary observer is :

(A) ωR (B) $2\omega R$ (C) $3\omega R$ (D) zero

12. A transverse periodic wave on a string with a linear mass density of 0.200 kg/m is described by the following equation $y = 0.05 \sin(420t - 21.0 x)$

where x and y are in metres and t is in seconds. The tension in the string is equal to :

(A) 32 N (B) 42 N (C) 66 N (D) 80 N

13. A wave moving with constant speed on a uniform string passes the point x = 0 with amplitude A_0, angular frequency ω_0 and average rate of energy transfer P_0. As the wave travels down the string it gradually loses energy and at the point $x = \ell$, the average rate of energy transfer becomes $\frac{P_0}{2}$. At the point $x = \ell$, angular frequency and amplitude are respectively :

(A) ω_0 and $A_0/\sqrt{2}$ (B) $\omega_0/\sqrt{2}$ and A_0

(C) less than ω_0 and A_0 (D) $\omega_0/\sqrt{2}$ and $A_0/\sqrt{2}$

14. A sinusoidal wave with amplitude y_m is travelling with speed V on a string with linear density ρ. The angular frequency of the wave is ω. The following conclusions are drawn. Mark the one which is correct.

(A) doubling the frequency doubles the rate at which energy is carried along the string

(B) if the amplitude were doubled, the rate at which energy is carried would be halved

(C) if the amplitude were doubled, the rate at which energy is carried would be doubled

(D) the rate at which energy is carried is directly proportional to the velocity of the wave.

15. Sinusoidal waves 5.00 cm in amplitude are to be transmitted along a string having a linear mass density equal to 4.00×10^{-2} kg/m. If the source can deliver a average power of 90 W and the string is under a tension of 100 N, then the highest frequency at which the source can operate is (take $\pi^2 = 10$):

(A) 45 Hz (B) 50 Hz (C) 30 Hz (D) 62 Hz

16. The average power transmitted through a given point on a string supporting a sine wave is. 0.40 watt when the amplitude of wave is 2 mm. What average power will transmitted through this point its amplitude is increased to 4 mm.

(A) 0.40 watt (B) 0.80 watt (C) 1.2 watt (D) 1.6 watt

Power of real gurus

17. When two waves of the same amplitude and frequency but having a phase difference of ϕ, travelling with the same speed in the same direction (positive x), meets at a point then.

(A) their resultant amplitude will be twice that of a single wave but the frequency will be same

(B) their resultant amplitude and frequency will both be twice that of a single wave

(C) their resultant amplitude will depend on the phase angle while the frequency will be the same

(D) the frequency and amplitude of the resultant wave will depend upon the phase angle.

18. The rate of transfer of energy in a wave depends

(A) directly on the square of the wave amplitude and square of the wave frequency

(B) directly on the square of the wave amplitude and square root of the wave frequency

(C) directly on the wave frequency and square of the wave amplitude

(D) directly on the wave amplitude and square of the wave frequency

19. Two waves of amplitude A_1, and A_2 respectively and equal frequency travels towards same point. The amplitude of the resultant wave is

(A) $A_1 + A_2$ (B) $A_1 - A_2$

(C) between $A_1 + A_2$ and $A_1 - A_2$ (D) Can not say

20. A wave pulse, travelling on a two piece string, gets partially reflected and partially transmitted at the junction. The reflected wave is inverted in shape as compared to the incident one. If the incident wave has speed v and the transmitted wave v′,

(A) $v' > v$

(B) $v' = v$

(C) $v' < v$

(D) nothing can be said about the relation of v and v′.

21. The effects are produced at a given point in space by two waves described by the equations, $y_1 = y_m \sin \omega t$ and $y_2 = y_m \sin (\omega t + \phi)$ where y_m is the same for both the waves and ϕ is a phase angle. Tick the incorrect statement among the following.

(A) the maximum intensity that can be achieved at a point is twice the intensity of either wave and occurs if $\phi = 0$

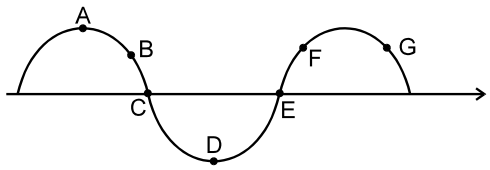
(B) the maximum intensity that can be achieved at a point is four times the intensity of either wave and occurs if $\phi = 0$

(C) the maximum amplitude that can be achieved at the point its twice the amplitude of either wave and occurs at $\phi = 0$

(D) When the intensity is zero, the net amplitude is zero, and at this point $\phi = \pi$.

Power of real gurus

22. The following figure depicts a wave travelling in a medium. Which pair of particles are in phase.



(A) A and D (B) B and F (C) C and E (D) B and G

23. A wave represented by the equation $y = a\cos(kx - \omega t)$ is superposed with another wave to form a stationary wave such that the point $x = 0$ is a node . The equation for other wave is :

(A) $a\sin(kx + \omega t)$ (B) $-a\cos(kx + \omega t)$ (C) $-a\cos(kx - \omega t)$ (D) $-a\sin(kx - \omega t)$

24. Two stretched wires A and B of the same lengths vibrate independently. If the radius, density and tension of wire A are respectively twice those of wire B, then the fundamental frequency of vibration of A relative to that of B is

(A) 1 : 1 (B) 1 : 2 (C) 1 : 4 (D) 1 : 8

25. A steel wire of mass 4.0 g and length 80 cm is fixed at the two ends. The tension in the wire is 50 N. The wavelength of the fourth harmonic of the fundamental will be :

(A) 80 cm (B) 60 cm (C) 40 cm (D) 20 cm

Power of real gurus

ANSWER KEY

1.	(D)	**2.**	(A)	**3.**	(C)	**4.**	(B)	**5.**	(B)
6.	(B)	**7.**	(C)	**8.**	(B)	**9.**	(B)	**10.**	(A)
11.	(B)	**12.**	(D)	**13.**	(A)	**14.**	(D)	**15.**	(C)
16.	(D)	**17.**	(C)	**18.**	(A)	**19.**	(C)	**20.**	(C)
21.	(A)	**22.**	(D)	**23.**	(B)	**24.**	(B)	**25.**	(C)